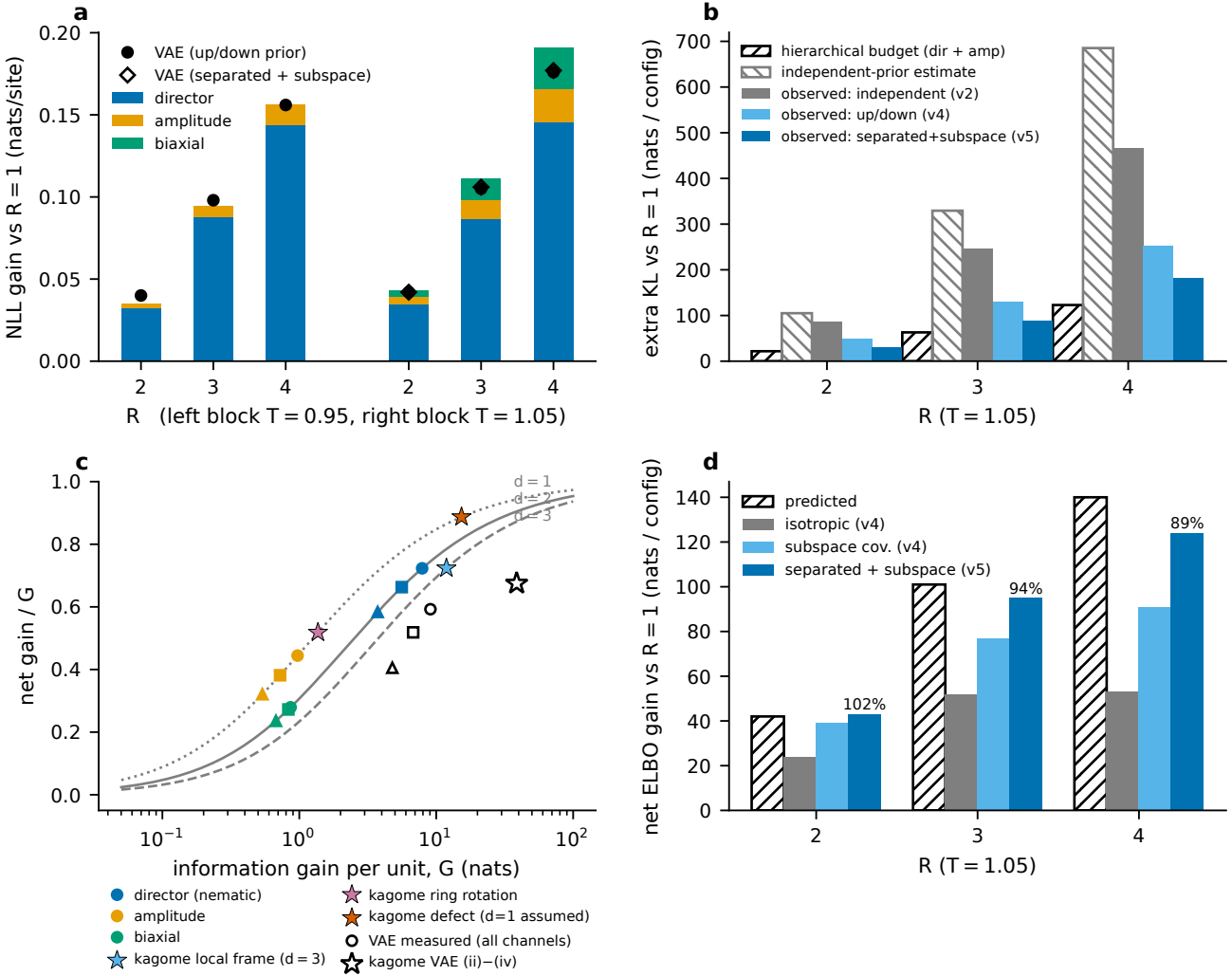
a
NLL gain vs R = 1 (nats/site)
VAE (up/down prior)
VAE (separated + subspace)
director
amplitude
biaxial
R (left block T = 0.95, right block T = 1.05)
b
extra KL vs R = 1 (nats / config)
hierarchical budget (dir + amp)
independent-prior estimate
observed: independent (v2)
observed: up/down (v4)
observed: separated+subspace (v5)
R (T = 1.05)
c
net gain / G
d = 1
d = 2
d = 3
information gain per unit, G (nats)
director (nematic)
amplitude
biaxial
kagome local frame (d = 3)
kagome ring rotation
kagome defect (d=1 assumed)
VAE measured (all channels)
kagome VAE (ii)−(iv)
d
net ELBO gain vs R = 1 (nats / config)
predicted
isotropic (v4)
subspace cov. (v4)
separated + subspace (v5)
102%
94%
89%
R (T = 1.05)